$$\hat{\rho}_{\mathcal{S}}(t) = \mathrm{Tr}_{\mathcal{E}} \left|\Psi(t)\right\rangle\!\left\langle\Psi(t)\right| = \sum_{n=0}^{2^N-1} \left|c_n\right|^2 \hat{U}_n(t)\hat{\rho}_{\mathcal{S}}(0)\hat{U}_n^{\dagger}(t), \tag{5.158}$$